

Newton's Laws: Worked Solutions

4.33–4.49 odd • Nine problems • September 22, 2026

Use $g = 9.80 \text{ m/s}^2$. Questions transcribed or closely paraphrased from your screenshots.

4.33 | Bucket and breaking strength

Question. A 5.60 kg bucket of water is accelerated upward by a cord of negligible mass whose breaking strength is 75.0 N. If the bucket starts from rest, what is the minimum time required to raise the bucket a vertical distance of 12.0 m without breaking the cord?

Forces and maximum acceleration. Take upward as positive. The cord pulls upward with tension T , and gravity pulls downward with weight mg . The fastest rise uses the largest allowed tension throughout: $T = 75.0 \text{ N}$.

$$T - mg = ma$$

$$a_{\text{max}} = (75.0 - 5.60 \times 9.80)/5.60 = 3.59286 \text{ m/s}^2.$$

Time from rest. $\Delta y = \frac{1}{2}at^2$, so $t_{\text{min}} = \sqrt{(2\Delta y/a_{\text{max}})}$
 $= \sqrt{(2 \times 12.0/3.59286)} = 2.58455 \text{ s}.$

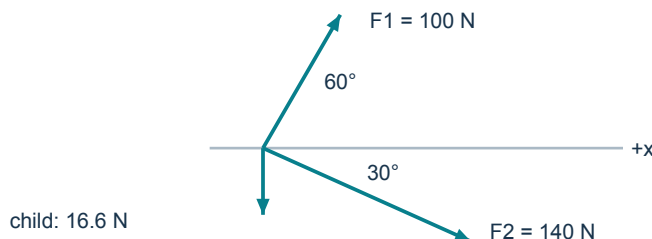
Answer. $t_{\text{min}} = 2.58 \text{ s}$. This is the first time the bucket reaches 12.0 m; the question does not require it to be at rest at that height.

4.35 | Smallest push on a cart

Question. Two adults and a child want to push a wheeled cart in the direction marked x in Fig. P4.35. The two adults push with horizontal forces F_1 and F_2 as shown. (a) Find the magnitude and direction of the smallest force that the child should exert. Ignore friction. (b) If the child exerts that minimum force, the cart accelerates at 2.0 m/s^2 in the $+x$ -direction. What is the weight of the cart?

Figure P4.35: top view. $F_1 = 100 \text{ N}$ at 60° above $+x$; $F_2 = 140 \text{ N}$ at 30° below $+x$. Both forces lie in the horizontal plane.

Top view



(a) Cancel the sideways component. $F_{\text{adults},y} = 100 \sin 60^\circ - 140 \sin 30^\circ = +16.6025 \text{ N}.$

Thus $F_{\text{child},y} = -16.6025 \text{ N}$. Since $|F_{\text{child}}| = \sqrt{(F_{\text{child},x}^2 + F_{\text{child},y}^2)}$, its magnitude is smallest when $F_{\text{child},x} = 0$.

Answer (a). 16.6 N in the $-y$ -direction, perpendicular to $+x$ (downward on the top-view diagram, not toward the ground).

(b) Find mass, then weight. $F_{\text{net},x} = 100 \cos 60^\circ + 140 \cos 30^\circ = 171.244 \text{ N}.$

$$m = F_{\text{net},x}/a = 171.244/2.0 = 85.6218 \text{ kg}.$$

$$W = mg = 85.6218 \times 9.80 = 839.093 \text{ N}.$$

Answer (b). $W \approx 8.4 \times 10^2 \text{ N}$. Weight is a force, so the answer is in newtons.

4.37 | Two crates connected by a rope

Question. Two crates, one of mass 4.00 kg and the other of mass 6.00 kg, sit on the frictionless surface of a frozen pond, connected by a light rope (Fig. P4.37). A woman wearing golf shoes for traction pulls horizontally on the 6.00 kg crate with a force F that gives it an acceleration of 2.50 m/s^2 . (a) What is the acceleration of the 4.00 kg crate? (b) Draw its free-body diagram and use Newton's second law to find the connecting-rope tension T . (c) Draw the free-body diagram for the 6.00 kg crate. What is the direction of its net force? Which is larger, T or F ? (d) Use part (c) and Newton's second law to calculate F .

Figure P4.37: 4.00 kg crate on the left, rope, 6.00 kg crate on the right; the woman pulls the 6.00 kg crate to the right.



(a) Shared acceleration. The taut, nonstretching rope keeps their separation fixed. **Both crates accelerate at 2.50 m/s^2 to the right.**

(b) Isolate the 4.00 kg crate. Vertical forces balance: $N_4 = 4.00g = 39.2 \text{ N}$. Horizontally, $T = m_4a = 4.00 \times 2.50$. **$T = 10.0 \text{ N}$.**

(c) Isolate the 6.00 kg crate. Vertical forces balance: $N_6 = 6.00g = 58.8 \text{ N}$. Horizontally, $F - T = m_6a > 0$. **The net force points right, and $F > T$.**

(d) Applied force. $F = T + m_6a = 10.0 + 6.00 \times 2.50 = 25.0 \text{ N to the right}$.
Check using both crates: $F = (4.00 + 6.00) \times 2.50 = 25.0 \text{ N}$.

4.39 | Test gun with a changing acceleration

Question. To study damage to aircraft that collide with large birds, you design a test gun that accelerates chicken-sized objects so that their displacement along the barrel is $x(t) = (9.0 \times 10^3 \text{ m/s}^2)t^2 - (8.0 \times 10^4 \text{ m/s}^3)t^3$. The object leaves the end of the barrel at $t = 0.025 \text{ s}$. (a) How long must the barrel be? (b) What is the speed as the object leaves? (c) What net force must be exerted on a 1.50 kg object at (i) $t = 0$ and (ii) $t = 0.025 \text{ s}$?

(a) Barrel length. $x(0.025) = 9000(0.025)^2 - 80000(0.025)^3$
 $= 5.625 - 1.250 = 4.375 \text{ m}$. **Length $\approx 4.4 \text{ m}$.**

(b) Differentiate once. $v(t) = dx/dt = (1.8 \times 10^4 \text{ m/s}^2)t - (2.4 \times 10^5 \text{ m/s}^3)t^2$.
 $v(0.025) = 450 - 150 = 3.0 \times 10^2 \text{ m/s}$.

(c) Differentiate again and apply $F_{\text{net}} = ma$. $a(t) = (1.8 \times 10^4 \text{ m/s}^2) - (4.8 \times 10^5 \text{ m/s}^3)t$.
(i) $F_{\text{net}}(0) = 1.50 \times 18000 = 2.7 \times 10^4 \text{ N}$.
(ii) $a(0.025) = 18000 - 12000 = 6000 \text{ m/s}^2$,
so $F_{\text{net}}(0.025) = 1.50 \times 6000 = 9.0 \times 10^3 \text{ N}$.
Both net forces point along $+x$, toward the barrel's exit.

4.41 | Elevator scale readings

Question. After an annual checkup, you leave your physician's office, where you weighed 683 N. You then get into an elevator that conveniently has a scale. Find the magnitude and direction of the elevator's acceleration if the scale reads (a) 725 N and (b) 595 N.

A scale measures the normal force. Your mass is $m = W/g = 683/9.80 = 69.6939$ kg. Take upward as positive. The scale pushes upward with normal force N , while your weight W acts downward.
 $N - W = ma$, so $a = (N - W)/m$.

(a) Reading 725 N. $a = (725 - 683)/69.6939 = +0.602635$ m/s².

Acceleration: 0.603 m/s² upward.

(b) Reading 595 N. $a = (595 - 683)/69.6939 = -1.26266$ m/s².

Acceleration: 1.26 m/s² downward.

Interpretation. A reading above your true weight means upward acceleration; a reading below it means downward acceleration. The readings alone do not tell whether the elevator is moving up or down.

4.43 | Bat reverses a baseball's velocity

Question. A batter swings at a baseball of mass 0.145 kg moving horizontally toward him at 40.0 m/s. He hits a line drive, with the ball moving away from him horizontally at 50.0 m/s just after it leaves the bat. If the bat and ball are in contact for 8.00 ms, what is the average force that the bat applies to the ball?

Choose the outgoing direction as positive. $v_i = -40.0$ m/s, $v_f = +50.0$ m/s.

$\Delta v = v_f - v_i = 50.0 - (-40.0) = 90.0$ m/s.

$\Delta t = 8.00$ ms = 8.00×10^{-3} s.

Average horizontal force. $F_{\text{avg},x} = m\Delta v/\Delta t$
 $= (0.145 \times 90.0)/(8.00 \times 10^{-3}) = 1631.25$ N.

Answer. 1.63×10^3 N away from the batter (in the outgoing horizontal direction, to the stated precision).

Why the speeds add. The ball reverses direction, so its velocity changes by 90.0 m/s, not 10.0 m/s. The average force is determined by the total velocity change; the force need not be constant.

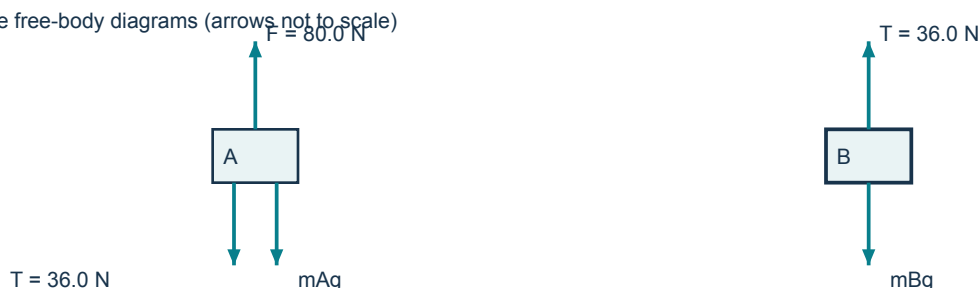
Gravity detail. If the initial and final velocities are exactly horizontal, the bat also supplies an average upward component $mg = 1.42$ N to cancel gravity's vertical impulse. This changes neither the rounded magnitude nor the nearly horizontal direction above.

4.45 | Two descending boxes pulled upward

Question. Boxes A and B are connected to each end of a light vertical rope (Fig. P4.45). A constant upward force $F = 80.0$ N is applied to box A. Starting from rest, box B descends 12.0 m in 4.00 s. The tension in the rope connecting the two boxes is 36.0 N. What are the masses of (a) box B and (b) box A?

Figure P4.45: box A above box B, joined by a vertical rope; the external 80.0 N force acts upward on A.

Separate free-body diagrams (arrows not to scale)



Get the acceleration from the motion. Choose downward as positive. Both boxes have the same constant acceleration. $12.0 = \frac{1}{2}a(4.00)^2$, so $a = 1.50 \text{ m/s}^2$ downward.

(a) Lower box B. Forces: $m_B g$ down and T up.

$$m_B g - T = m_B a \rightarrow m_B = T/(g - a).$$

$$m_B = 36.0/(9.80 - 1.50) = 4.34 \text{ kg}.$$

(b) Upper box A. Forces: $m_A g$ and T down; F up.

$$m_A g + T - F = m_A a \rightarrow m_A = (F - T)/(g - a).$$

$$m_A = (80.0 - 36.0)/(9.80 - 1.50) = 5.30 \text{ kg}.$$

System check. $(m_A + m_B)(g - a) = 80.0$ N. The internal tension cancels when both boxes are treated as one system.

4.47 | Rocket makes a soft landing

Question. A small rocket of mass 20.0 kg is in free fall toward Earth. Air resistance is negligible. At 80.0 m above the surface, it is moving downward at 30.0 m/s. Its engines then produce a constant upward force F . Neglect the change in mass. What value of F makes its speed zero just as it reaches Earth's surface, for a soft landing? Hint: the net force combines the upward engine force and the downward weight.

Find the required acceleration. Choose upward as positive: $v_i = -30.0$ m/s, $v_f = 0$, $\Delta y = -80.0$ m.

$$v_f^2 = v_i^2 + 2a\Delta y \rightarrow 0 = 900 - 160a.$$

$$a = +5.625 \text{ m/s}^2 \text{ (upward)}.$$

Include gravity when finding thrust. $F - mg = ma \rightarrow F = m(g + a).$

$$F = 20.0(9.80 + 5.625) = 308.5 \text{ N}.$$

Answer. $F = 309$ N upward. The upward net force is only $ma = 112.5$ N; the engines must also overcome the 196 N weight.

Check. $t = (0 - (-30.0))/5.625 = 5.33$ s, and $\Delta y = \frac{1}{2}(v_i + v_f)t = -80.0$ m. Upward acceleration slows the downward motion to rest at the surface.

4.49 | Rocket force increases with time

Question. A mysterious rocket-propelled object of mass 45.0 kg is initially at rest in the middle of the horizontal, frictionless surface of an ice-covered lake. A force directed east with magnitude $F(t) = (16.8 \text{ N/s})t$ is applied. How far does the object travel in the first 5.00 s after the force is applied?

Acceleration is not constant. Choose east as positive and set $x(0) = 0$. Vertically, the surface's normal force balances the weight. The applied force is the net horizontal force.

$$a(t) = F(t)/m = [(16.8 \text{ N/s})/(45.0 \text{ kg})]t = (0.373333 \text{ m/s}^3)t.$$

Integrate to get velocity. Using $v(0) = 0$:

$$\begin{aligned} v(t) &= \int_0^t a(\tau) d\tau = (16.8/(2 \times 45.0))t^2 \\ &= (0.186667 \text{ m/s}^3)t^2. \end{aligned}$$

Integrate again to get displacement. Using $x(0) = 0$:

$$\begin{aligned} x(t) &= \int_0^t v(\tau) d\tau = (16.8/(6 \times 45.0))t^3. \\ x(5.00) &= (16.8 \times 5.00^3)/(6 \times 45.0) = 7.77778 \text{ m}. \end{aligned}$$

Answer. 7.78 m east. The velocity stays nonnegative, so distance traveled equals the magnitude of displacement.

Common trap. Do not use the final acceleration in $x = \frac{1}{2}at^2$: that formula requires constant acceleration, but here a grows linearly with time.

Quick answer check

4.33 2.58 s

4.35 (a) 16.6 N along $-y$; (b) 8.4×10^2 N

4.37 (a) 2.50 m/s^2 right; (b) 10.0 N; (c) net force right, $F > T$; (d) 25.0 N

4.39 (a) 4.4 m; (b) $3.0 \times 10^2 \text{ m/s}$; (c) 2.7×10^4 N and 9.0×10^3 N

4.41 (a) 0.603 m/s^2 up; (b) 1.26 m/s^2 down

4.43 1.63×10^3 N, away from the batter

4.45 (a) $m_B = 4.34 \text{ kg}$; (b) $m_A = 5.30 \text{ kg}$

4.47 309 N upward

4.49 7.78 m east

Source: three user-supplied textbook screenshots, September 22, 2026, including page 126 and Figures P4.35, P4.37, and P4.45. Textbook title and edition are not established by these images.